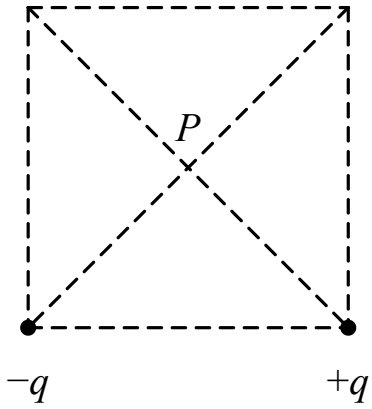
STPM 2025 PHYSICS 960/2

Questions and Answers

Values of constants: $e = 1.60 \times 10^{-19}\ C$, $m_p = 1.67 \times 10^{-27}\ kg$, $\varepsilon_0 = 8.85 \times 10^{-12}\ F\ m^{-1}$, $\mu_0 = 4\pi \times 10^{-7}\ H\ m^{-1}$, $N_A = 6.02 \times 10^{23}\ mol^{-1}$.

Section A [15 marks]

1. Two isolated point charges, $-q$ and $+q$, are located at two vertices of a square as shown in the diagram below.



Which is the direction of the net electric field at P which lies on the same plane as the charges?

A ↑ B ↓ C ← D →

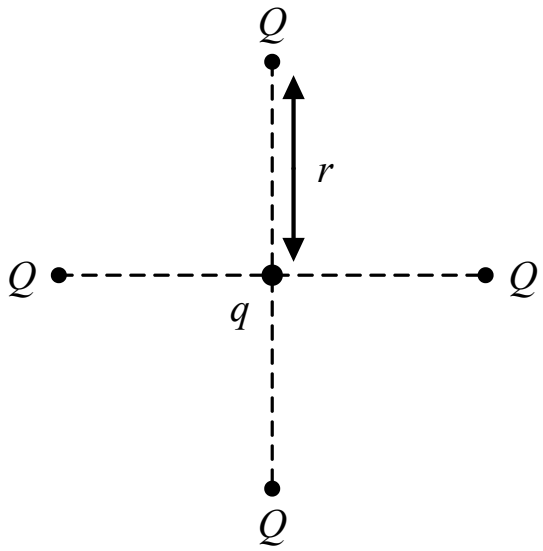
Answer: C

Field of $+q$ at P points away from $+q$ (up and to the left); field of $-q$ points towards $-q$ (down and to the left).

The vertical components cancel and the horizontal components add: the net field points to the left (←).

Explanation: Draw each field as a vector from P; both charges are the same distance from P, so the two fields have equal magnitudes.

2. A point charge, q, is located at an equidistant, r, from four identical point charges, Q, as shown in the diagram below.



Which is the work done to bring q to infinity?

A $\frac{4Qq}{\pi\varepsilon_o r}$ B $\frac{Qq}{4\pi\varepsilon_o r}$ C $\frac{Qq}{2\pi\varepsilon_o r}$ D $\frac{Qq}{\pi\varepsilon_o r}$

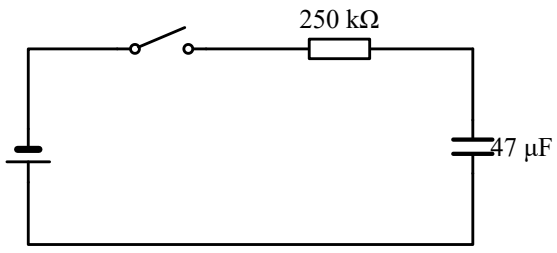
Answer: D

Potential at q: $V = 4 \times \frac{Q}{4\pi\varepsilon_o r} = \frac{Q}{\pi\varepsilon_o r}$

Work done (by the field) moving q to infinity $= qV = \frac{Qq}{\pi\varepsilon_o r}$

Explanation: Potential is a scalar, so the four contributions simply add.

3. A capacitor, a resistor, a switch and a battery are connected in series as shown in the circuit diagram below.



What is the time taken for the capacitor to reach 63% of the supply voltage after the switch is closed?

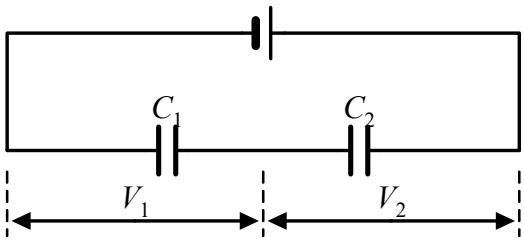
A 3.8 s B 5.5 s C 8.2 s D 11.8 s

Answer: D

Time constant $\tau = RC = (250 \times 10^3)(47 \times 10^{-6}) = 11.8$ s

Explanation: After one time constant a charging capacitor reaches $(1 - e^{-1}) = 63\%$ of the supply voltage.

4. Two capacitors, C_1 and C_2, with the capacitance of 2 μF and 3 μF respectively are connected in series to an e.m.f. source as shown in the circuit diagram below.



Q_1 and Q_2 are charges stored in C_1 and C_2 respectively. Which potential differences, V_1 and V_2, correspond correctly to Q_1 and Q_2?

	Potential differences	*Charges stored*
A	$V_1 > V_2$	$Q_1 < Q_2$
B	$V_1 > V_2$	$Q_1 = Q_2$
C	$V_1 < V_2$	$Q_1 = Q_2$
D	$V_1 < V_2$	$Q_1 < Q_2$

Answer: B

In series both capacitors carry the same charge: $Q_1 = Q_2$.

$V = \frac{Q}{C}$: the smaller capacitor ($C_1 = 2$ μF) has the larger p.d., $V_1 > V_2$

Explanation: Charge flows onto one plate of C_1 only as fast as it leaves the facing plate of C_2.

5. Which statement is true about drift velocity of free electrons in a metal conductor?

A The free electrons drift in the opposite direction of the electric field applied.
B The direction of the drift velocity is the same as the direction of the current flow.
C The number of free electrons that drifts is equal to the number of positive charges.
D The drift velocity of the free electrons is the same as the net velocity in random motion.

Answer: A

Electrons are negative, so they drift opposite to the applied electric field (and opposite to the conventional current).

Explanation: The drift velocity is tiny compared with the random thermal speed of the electrons.

6. Copper has a density of 8950 kg m^{-3} and molar mass of 63.5 g mol^{-1}. A copper wire of cross-sectional area 3.8×10^{-6} m^2 is used as a conductor in a circuit. If each copper atom provides one free electron for conduction, what is the drift velocity of electrons in the wire when the wire carries a current of 1.5 A?

A 7.2×10^{-6} m s^{-1} B 2.9×10^{-5} m s^{-1} C 7.2×10^{-3} m s^{-1} D 2.9×10^{-2} m s^{-1}

Answer: B

$$n = \frac{8950}{0.0635} \times 6.02 \times 10^{23} = 8.48 \times 10^{28}\ \text{m}^{-3}$$

$$v_d = \frac{I}{nAe} = \frac{1.5}{(8.48 \times 10^{28})(3.8 \times 10^{-6})(1.60 \times 10^{-19})} = 2.9 \times 10^{-5}\ \text{m s}^{-1}$$

Explanation: Number of atoms per m^3 = (density ÷ molar mass) × N_A, with molar mass in kg mol^{-1}.

7. One helium ion, He^{2+}, moves with a speed of 5.50×10^6 m s^{-1} in a magnetic field of 2.00 T experiences a magnetic force of 9.20×10^{-13} N. What is the angle between the path of He^{2+} and the direction of the magnetic field?

A 15.2° B 31.5° C 58.5° D 74.8°

Answer: A

$$F = qvB \sin\theta \text{ with } q = 2e\text{: } \sin\theta = \frac{9.20 \times 10^{-13}}{(3.20 \times 10^{-19})(5.50 \times 10^{6})(2.00)} = 0.261 \Longrightarrow \theta = 15.2°$$

Explanation: He^{2+} carries a charge of $2e$.

8. A circular coil of radius 15 cm with 100 turns is placed in a uniform magnetic field which has a flux density 0.75 T. If the angle between the plane of the coil and the magnetic field is 60°, what is the magnetic flux linkage through the coil?

A 0 Wb B 2.7 Wb C 4.6 Wb D 5.3 Wb

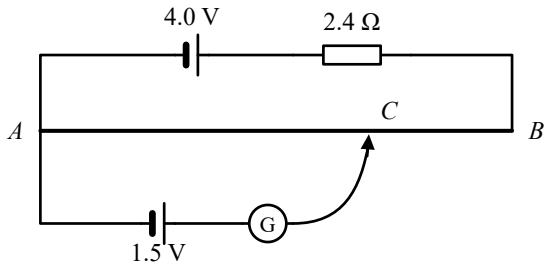
Answer: C

The plane is at 60° to **B**, so the normal makes 30° with **B**.

$$N\Phi = NBA\cos 30° = (100)(0.75)\pi(0.15)^2 \cos 30° = 4.6\ \text{Wb}$$

Explanation: Flux uses the angle between **B** and the normal to the coil.

9. A potentiometer with a slide wire, AB, of length 100 cm and resistance 2.0 Ω is connected to a resistor of resistance 2.4 Ω as shown in the circuit diagram below.



The electromotive forces, e.m.f., of the driver cell and the cell are 4.0 V and 1.5 V respectively. If the internal resistance of the driver cell is negligible, what is the balanced length of AC?

A 18 cm B 38 cm C 69 cm D 83 cm

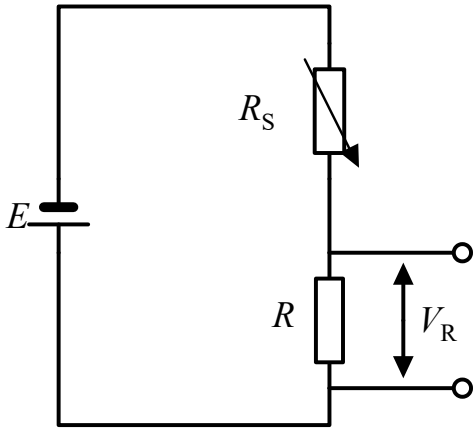
Answer: D

$$\text{Current in the wire} = \frac{4.0}{2.4 + 2.0} = 0.909 \text{ A; p.d. across } AB = (0.909)(2.0) = 1.82 \text{ V}$$

$$\text{p.d. per cm} = 0.0182 \text{ V cm}^{-1} \Longrightarrow AC = \frac{1.5}{0.0182} = 83 \text{ cm}$$

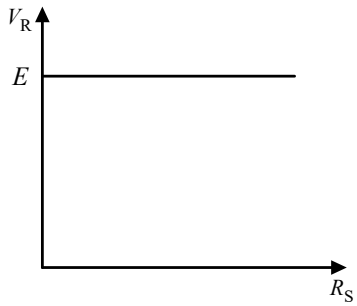
Explanation: At balance no current flows in the galvanometer, so the cell's e.m.f. equals the p.d. across *AC*.

10. A variable resistor, R_S, and a resistor, R, are connected in series to a potential supply, E, as shown in the circuit diagram below.

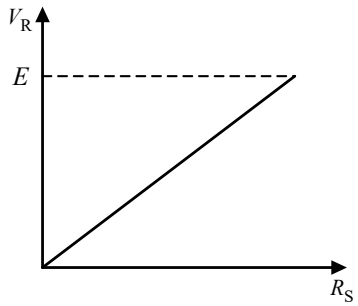


Which graph represents the variation of output voltage, V_R, with R_S for the circuit?

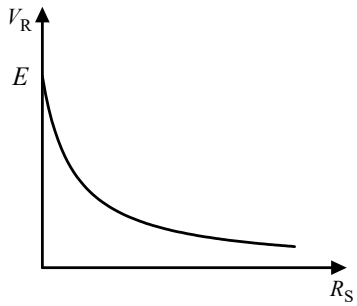
A



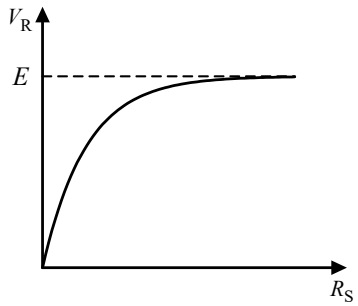
B



C



D

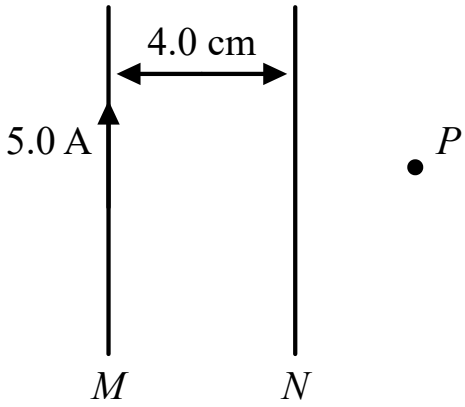


Answer: C

$V_R = \frac{ER}{R + R_S}$: equal to E when $R_S = 0$, then decreasing towards zero (C)

Explanation: A potential divider: the larger R_S, the bigger its share of E.

11. Two identical parallel wires, M and N, are placed at 4.0 cm from each other as shown in the diagram below.



As the current of 5.0 A flows through M and an unknown current flows through N, the force per unit length 7.5×10^{-5} N m^{-1} attracts the wires to each other. Which are the magnetic field strength of N at M and the direction of the magnetic field at point P?

	Magnetic field strength at M	*Direction of magnetic field at point P*
A	1.5×10^{-5} T	Into the page
B	1.5×10^{-5} T	Out of the page
C	2.5×10^{-5} T	Into the page
D	2.5×10^{-5} T	Out of the page

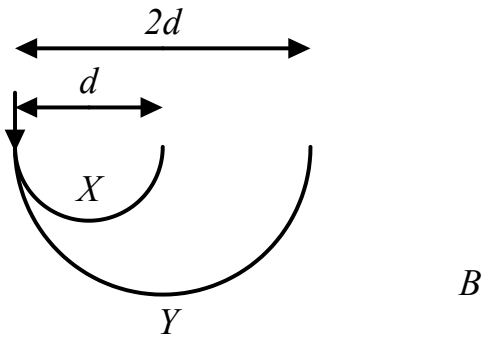
Answer: A

$$F/l = BI_M \Longrightarrow B = \frac{7.5 \times 10^{-5}}{5.0} = 1.5 \times 10^{-5} \text{ T}$$

Attraction $\Longrightarrow$ both currents flow upward. To the right of the wires each upward current produces a field into the page at P.

Explanation: Use the right-hand grip rule for each wire.

12. Two charged particles, X and Y, enter perpendicularly into a uniform magnetic field, B, at the same speed as shown in the diagram below.



Which statement is true about X and Y?

A The charge of Y is larger than the charge of X.
B The mass of Y is larger than the mass of X.
C The ratio of $\frac{q}{m}$ of Y is two times the ratio of $\frac{q}{m}$ of X.
D The ratio of $\frac{q}{m}$ of Y is half the ratio of $\frac{q}{m}$ of X.

Answer: D

$$r = \frac{mv}{qB}\text{: same } v \text{ and } B,\ r_Y = 2r_X \Longrightarrow (q/m)_Y = \tfrac{1}{2}(q/m)_X$$

Explanation: Only the ratio q/m can be found, not the charge or mass separately.

13. A coil is placed in a magnetic field. Which is the effect of the changes of the magnetic field that passes through the coil?

A An induced e.m.f. in the coil causes the current flows and the induced current opposes the changes in the magnetic field.
B An induced e.m.f. causes the magnetic field produced in the coil and the induced current opposes the changes in the magnetic field.
C An induced e.m.f. in the coil causes the current flows and the induced current is in the direction of change in the magnetic field.
D An induced e.m.f. causes the magnetic field produced in the coil and the induced current is in the direction of change in the magnetic field.

Answer: A

A changing flux induces an e.m.f. (Faraday), which drives a current; the induced current opposes the change producing it (Lenz).

Explanation: The opposition is required by conservation of energy.

14. In an alternating current circuit, the mean power dissipated in a resistor of resistance 150 Ω is 16 W. What is the peak current of the resistor?

A 0.65 A B 0.46 A C 0.33 A D 0.23 A

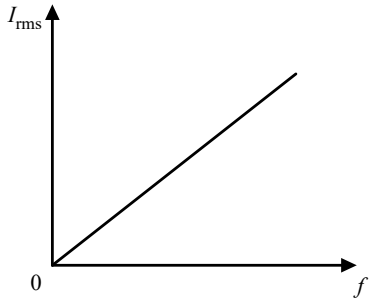
Answer: B

$$\langle P \rangle = \frac{I_0^2 R}{2} \Longrightarrow I_0 = \sqrt{\left(\frac{2(16)}{150}\right)} = 0.46 \text{ A}$$

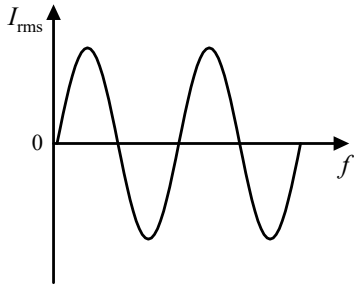
Explanation: Mean power uses the r.m.s. current, $I_0/\sqrt{2}$.

15. A pure inductor is connected in series to an alternating current power supply with a constant r.m.s. voltage and variable frequency, f. Which graph shows the variation of the r.m.s. current, I_{rms}, with f?

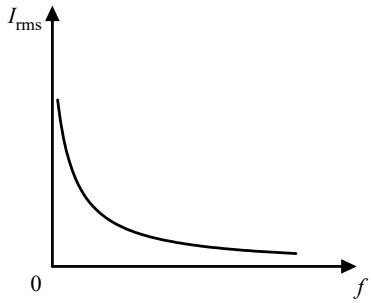
A



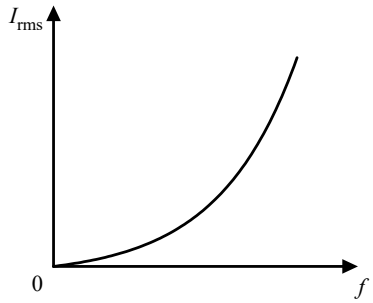
B



C



D



Answer: C

$$I_{rms} = \frac{V_{rms}}{X_L} = \frac{V_{rms}}{2\pi f L} \Longrightarrow I_{rms} \propto \frac{1}{f} \text{ (C)}$$

Explanation: An inductor opposes rapidly changing currents more.

Section B [15 marks]

16. A capacitor of capacitance 100 pF is made of two square metal plates of length, L. Both metal plates are separated at the distance of 0.20 mm.

(a) Determine the value of L for the metal plates. [3]

Answer

$C = \frac{\varepsilon_0 A}{d}$ with $A = L^2$ — 1

$L = \sqrt{(\frac{Cd}{\varepsilon_0})} = \sqrt{(\frac{(100 \times 10^{-12})(0.20 \times 10^{-3})}{8.85 \times 10^{-12}})}$ — 1

$L = 0.0475$ m — 1

3

Explanation: The plates are square, so the area is L^2.

(b) If the space between the plates of the capacitor is filled with a material of dielectric constant 5.6, determine the capacitance of the capacitor. [2]

Answer

$C' = \varepsilon_r C = (5.6)(100 \text{ pF})$ — 1

$= 560$ pF — 1

2

Explanation: A dielectric increases the capacitance by the factor ε_r.

(c) A battery is connected to fully charged the capacitor in (b) and then is disconnected. Explain the effect on the energy stored in the capacitor when the dielectric is removed from the capacitor. [3]

Answer

With the battery disconnected the charge Q stays constant — 1

Removing the dielectric reduces the capacitance — 1

$U = \frac{Q^2}{2C}$ increases (by 5.6 times); the extra energy comes from the work done pulling the dielectric out — 1

3

Total for question **8**

Explanation: If the battery stayed connected, V would be constant and the energy would decrease instead.

17. A solenoid consists of 500 turns, length of 10 cm and diameter of 2.0 cm is connected to an alternating power supply.

(a) State **two** causes of e.m.f. induced in the solenoid. [2]

Answer

A changing current in the solenoid itself (self-induction) 1

A changing magnetic flux linkage through the solenoid, e.g. a moving magnet or a changing external field 1

2

Explanation: Any change in flux linkage induces an e.m.f. (Faraday's law).

(b) Determine the self-inductance in the solenoid. [3]

Answer

$L = \frac{\mu_0 N^2 A}{l}$, $A = \pi(0.010)^2$ 1

$L = \frac{(4\pi \times 10^{-7})(500)^2 \pi(0.010)^2}{0.10}$ 1

$L = 9.87 \times 10^{-4}$ H 1

3

Explanation: Use the radius (1.0 cm), not the diameter.

(c) If the current that flows in the solenoid is 2.0 A, determine the energy stored. [2]

Answer

$U = \frac{1}{2}LI^2 = \frac{1}{2}(9.87 \times 10^{-4})(2.0)^2$ 1

$U = 1.97 \times 10^{-3}$ J 1

2

Total for question **7**

Explanation: The energy is stored in the magnetic field of the solenoid.

Section C [30 marks]

18. (a) State **two** principles of Kirchhoff's laws. [2]

Answer

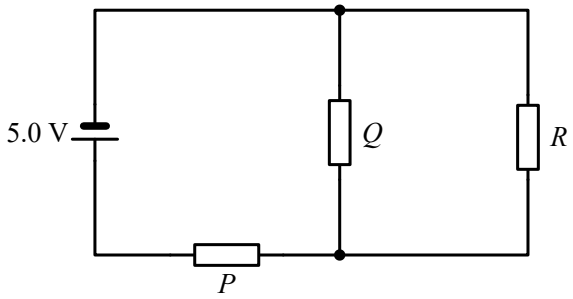
Junction rule: the sum of currents entering a junction equals the sum leaving (conservation of charge) 1

Loop rule: around any closed loop, the sum of e.m.f.s equals the sum of the p.d.s (IR) (conservation of energy) 1

2

Explanation: Kirchhoff's first and second laws.

(b) A battery of e.m.f. of 5.0 V is connected to three resistors, P, Q and R, of resistance 3.0 Ω, 5.0 Ω and 5.0 Ω respectively as shown in the circuit diagram below.



(i) Determine the effective resistance in the circuit. [3]

Answer

$Q \parallel R$: $\frac{(5.0)(5.0)}{5.0 + 5.0} = 2.5\ \Omega$	1
In series with P: 2.5 + 3.0	1
Effective resistance = 5.5 Ω	1
	3

Explanation: Q and R are in parallel; P is in the main line.

(ii) Determine the potential difference across R and then determine the current that flows through R. [5]

Answer

Total current $I = \frac{5.0}{5.5} = 0.909$ A	1
p.d. across the parallel part = $I(2.5)$	1
$V_R = 2.27$ V	1
$I_R = \frac{2.27}{5.0}$	1
$I_R = 0.455$ A	1
	5

Explanation: Equal resistors in parallel share the current equally.

(iii) If the total potential difference in the circuit is less than 5.0 V, state the cause of the difference. [1]

Answer

The battery has internal resistance, so some p.d. is lost inside the battery — 1

1

Explanation: Terminal p.d. = $E - Ir$.

(c) Explain the conversion process of a galvanometer to an ammeter. [4]

Answer

A low resistance (shunt) is connected in parallel with the galvanometer — 1

Most of the current passes through the shunt; only the full-scale current I_g passes through the galvanometer — 1

$I_g R_g = (I - I_g)R_s$ gives the shunt for a full-scale reading I — 1

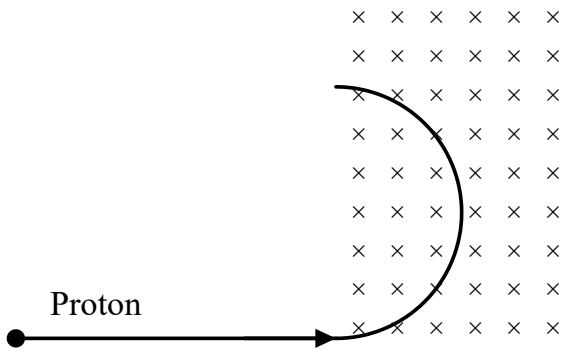
The ammeter then has a very low total resistance, so it hardly changes the current it measures — 1

4

Total for question **15**

Explanation: The scale is recalibrated to read the larger current.

19. (a) A proton enters perpendicularly to a uniform magnetic field of 3.0×10^{-3} T as shown in the diagram below.



(i) Calculate the time taken for the proton to complete a semicircular path. [7]

Answer

The magnetic force provides the centripetal force: $evB = \frac{mv^2}{r}$	1
$r = \frac{mv}{eB}$	1
Period $T = \frac{2\pi r}{v}$	1
$T = \frac{2\pi m}{eB}$ (independent of speed)	1
Semicircle: $t = \frac{T}{2} = \frac{\pi m}{eB}$	1
$t = \frac{\pi(1.67 \times 10^{-27})}{(1.60 \times 10^{-19})(3.0 \times 10^{-3})}$	1
$t = 1.09 \times 10^{-5}$ s	1
	7

Explanation: The speed is not needed: faster protons travel in bigger circles but take the same time.

(ii) Determine the radius of the circular path if the speed of the proton is 4.6×10^6 m s^{-1}. [1]

Answer

$r = \frac{mv}{eB} = \frac{(1.67 \times 10^{-27})(4.6 \times 10^6)}{(1.60 \times 10^{-19})(3.0 \times 10^{-3})} = 16.0$ m	1
	1

Explanation: A weak field gives a very large radius.

(b)

(i) Explain the Hall effect. [5]

Answer

A current flows through a conductor (or semiconductor) placed in a magnetic field perpendicular to the current	1
The moving charge carriers experience a magnetic force qvB and are pushed to one side	1
Charges build up on opposite faces, creating a transverse electric field and a p.d., the Hall voltage	1
Equilibrium is reached when the electric force balances the magnetic force: $qE = qvB$	1
$V_H = \frac{BI}{nqt}$	1
	5

Explanation: The sign of the Hall voltage shows whether the charge carriers are positive or negative.

(ii) State **two** applications of the Hall effect. [2]

Answer

• Hall probe to measure magnetic flux density	1
• Determining the type and number density of charge carriers in semiconductors / Hall-effect sensors (e.g. speed or position sensors)	1
	2
Total for question	**15**

Explanation: Any two applications.

20. (a) State a difference between a direct current and alternating current. [1]

Answer

Direct current flows in one direction only; alternating current reverses its direction periodically	1
	1

Explanation: Its magnitude also varies sinusoidally.

(b) A capacitor of capacitance 10 μF is connected to an alternating current supply with the voltage of $V = V_o \sin \omega t$. The frequency of the supply is 50 Hz.

(i) Derive an expression for a current that flows through the capacitor in terms of current, I, frequency, f, and time, t. [5]

Answer

$Q = CV = CV_o \sin \omega t$	1
$I = \frac{dQ}{dt}$	1
$= \omega CV_o \cos \omega t$	1
$\omega = 2\pi f$	1
$I = I_o \cos (2\pi ft)$, where $I_o = 2\pi fCV_o$	1
	5

Explanation: The current is the rate of flow of charge onto the plates.

(ii) State the phase difference between voltage and current that flow through the capacitor. [2]

Answer

The current leads the voltage — 1

by $\frac{\pi}{2}$ rad (90°) — 1

2

Explanation: cos ωt = sin ($\omega t + \pi/2$).

(iii) If the peak voltage is 2.0 V, determine the rate of energy stored in the capacitor at 3.0 s. [2]

Answer

$P = VI = V_o I_o \sin \omega t \cos \omega t = \frac{1}{2} V_o I_o \sin (4\pi ft)$ — 1

At t = 3.0 s: sin ($4\pi \times 50 \times 3.0$) = sin ($600\pi$) = 0, so the rate is 0 — 1

2

Explanation: Use radians on the calculator. Over each cycle the capacitor stores and returns energy, so the average power is zero.

(c) A pure inductor of inductance 55 mH and a resistor of resistance 25 Ω are connected in series to an alternating current supply of peak voltage 240 V and frequency of 50 Hz.

(i) Calculate the root mean square voltage of the circuit. [2]

Answer

$V_{rms} = \frac{V_0}{\sqrt{2}} = \frac{240}{\sqrt{2}}$ — 1

= 170 V — 1

2

Explanation: The r.m.s. value of a sinusoidal voltage is the peak value ÷ √2.

(ii) Determine the impedance of the circuit. [3]

Answer

$X_L = 2\pi fL = 2\pi(50)(55 \times 10^{-3}) = 17.3\ \Omega$ — 1

$Z = \sqrt{(R^2 + X_L^2)} = \sqrt{(25^2 + 17.3^2)}$ — 1

Z = 30.4 Ω — 1

3

Total for question **15**

Explanation: Resistance and reactance combine at right angles.